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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Second Semester B.Tech (CBCS) Degree Examinations

May 2025

EC1002-2 – APPLIED DIGITAL LOGIC DESIGN

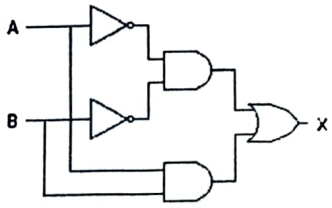
(For AC, EC, EE, VL, AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours | Max. Marks: 100

Note:
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.
Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III.**

PART - A: MULTIPLE CHOICE QUESTIONS — 20 Marks

1. **OR gate and------- will form the NOR gate?**
   A) OR gate B) NAND gate
   C) NOT gate D) AND gate
2. **Decimal equivalent of binary number 10101 is**
   A) 21 B) 12
   C) 22 D) 31
3. **A three variable karnaugh map has _____**
   A) Eight cells B) Three cells
   C) Sixteen Cells D) Four cells
4. **Which of the following is a SOP expression?**
   A) (A+B) (A+C) B) AB+AC
   C) Both a and b D) (A+B) (C+D)
5. **Which of the following logic expressions represents the logic diagram shown?**



   A) X=AB'+A'B B) X=(AB)'+AB
   C) X=(AB)'+A'B' D) X=A'B'+AB
6. **The decimal equivalent of the binary number 1011.011_2 is**
   A) 11.375_{10} B) 10.123_{10}
   C) 11.175_{10} D) 9.23_{10}
7. **Which of the following is not considered for forming groups in K-map?**
   A) Rolling B) Diagonal
   C) Vertical D) Horizontal
8. **The output of an exclusive-NOR gate is 1. Which input combination is correct?**
   A) $A = 1, B = 0$ B) $A = 0, B = 1$
   C) $A = 0, B = 0$ D) none of these
9. **How is JK flip-flop made to toggle**
   A) $J = 0, K = 0$ B) $J = 1, K = 0$
   C) $J = 1, K = 1$ D) $J = 0, K = 1$
10. **The characteristic equation for S-R flip flop is**
    A) $\bar{S} + \bar{R}Q$ B) $\bar{S} + R\bar{Q}$
    C) S+ RQ D) $S + \bar{R}Q$

11. **Full subtractor is used to perform subtraction of _____**
A) 2 bits B) 8 bits
C) 4 bits D) 3 bits

12. **The full form of SR is**
A) System rated B) Set reset
C) Set ready D) None of these

13. **A basic S-R flip-flop can be constructed by cross-coupling of which logic gates?**
A) AND or OR gates B) XOR or XNOR gates
C) NOR or NAND gates D) AND or NOR gates

14. **Latch is a device with _____ stable states**
A) One B) Three
C) Infinite D) Two

15. **The two input MUX would have _____**
A) 1 select line B) 2 select lines
C) 4 select lines D) 3 select lines

16. **Which of the following represents a number of output lines for a decoder with 4 input lines?**
A) 15 B) 16
C) 17 D) 18

17. **The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output shift register with an initial state 01110. After three clock pulses, the register contains _____.**
A) 01110 B) 00001
C) 00101 D) 10101

18. **A modulus-10 counter must have _____**
A) 4 flip-flops B) 5 flip-flops
C) 10 flip-flops D) 3 flip-flops

19. **In which one of the following counters, the flip flops are not clocked simultaneously?**
A) Synchronous counter B) Asynchronous counter/ripple counter
C) Both A and B D) None of these

20. **A shift register is a digital circuit that _____**
A) Stores data B) Shifts the data from left to right
C) Shifts the data from right to left D) All the above

PART - B: DESCRIPTIVE ANSWER QUESTIONS

		Unit – I	Marks	BT*	CO*	PO
1.	a)	Convert the following functions into canonical form. i) $f(w,x,y,z) = \bar{w}x + y\bar{z}$ (SOP) ii) $f(a,b,c,d) = (a + \bar{b} + c)(\bar{a} + d)$ (POS)	6	L*3	1	
	b)	Using K-map and simplify the following function i) $f(a,b,c) = \sum m(0,1,2,3,7)$ ii) $f(a,b,c,d) = \sum m(0,1,3,4,5,6,9,11,12,13,14)$	6	L3	2	
	c)	Define i) Minterm ii) Maxterm iii) Canonical SOP iv) Canonical POS	4	L2	1	
2.	a)	Design the combinational logic circuit which takes two, 2 bit numbers as its inputs and generates an output to indicate when the sum of the two numbers is odd.	6	L3	2	
	b)	i) $12.6532_{(10)} = (?)_{16}$ ii) $CF.52_{(16)} = (?)_{10}$ iii) $1376.185_{(10)} = (?)_2$	6	L2	1	
	c)	Perform the binary subtraction using 2's complement method. i) $11010010_2 - 01011110_2$ ii) $00110_2 - 11010_2$	4	L3	2	

a)	Design a combinational logic circuit to generate an output whenever a majority of four inputs is logic 1 and output function is not specified whenever the number of 0's and 1's is equal in the inputs. However, the output is logic zero for the remaining conditions.	8	L3	2	1
b)	Using the QM method simplify the following function $f(a,b,c,d) = \sum m\,(2,6,8,9,10,11,14,15)$	8	L3	2	1

Unit – II

a)	Implement $f(a,b,c) = \sum m(1,4,5,7)$ using i) 8:1 MUX ii) 4:1 MUX	6	L3	3	1
b)	With logic diagram and function table explain the operation of Set-Reset latch made with NOR gates.	6	L2	4	1
c)	Convert JK flip flop to T flip flop.	4	L3	4	1
a)	Explain the working of D flip flop with relevant logic diagram and truth table.	6	L2	4	1
b)	Implement the given function using 3 to 8 decoder with minimum number of gate inputs. i) $F1(a,b,c) = \sum m(0,1,3)$ ii) $F2(x,y,z) = \sum m(0,2,3,5,7)$ iii) $F3(p,q,r) = \pi M(1,2,3,5,6,7)$	6	L3	3	1
c)	Explain the working of 8 to 3 line Encoder	4	L2	3	1
6. a)	What are the characteristic equations of Flip flop? Obtain the characteristic equation for i) JK Flip flop ii) D Flip Flop	6	L3	4	1
b)	Explain the working of a 4-bit binary parallel adder/subtractor using XOR gates with a relevant block diagram and with an example	6	L2	3	1
c)	Design the full adder using 4:1 multiplexer.	4	L2	3	1

Unit – III

7. a)	Design a synchronous counter to sequence 0→4→1→2→6→0 using positive edge triggered D flip flops with minimal combinational gating.	8	L3	5	1
b)	What is a Ripple counter? Sketch the logic diagram of a 3-bit ripple-up counter using asynchronous positive edge triggered T flip flops. Explain its working with the sequence table and output waveforms.	8	L2	5	1
8. a)	Design a synchronous counter to sequence 0→2→3→6→5→1→0 using positive edge triggered T flip flops with minimal combinational gating.	8	L3	5	1
b)	What is a shift register? With the help of a suitable example explain the following operations: i) SIPO ii) PISO	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

First / Second Semester B.Tech (CBCS) Degree Examinations

Summer Semester July 2024

EC1002-2 – APPLIED DIGITAL LOGIC DESIGN

(For ACT,AD, AM, CB, CC, CS, IS, RI, EE, EC, VLSI)

ration: 3 Hours Max. Marks: 100

te:

rt – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each estion carries equal marks.

rt – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions each m **Unit – I & Unit – II each** and **One full** question from **Unit – III.**

PART - A: MULTIPLE CHOICE QUESTIONS **20 Marks**

---------- are universal logic gates.

A) NAND and NOR B) NOR AND EX-OR

C) AND and NOT D) OR and EX-OR

The base of hexadecimal number system is

A) 2 B) 8

C) 10 D) 16

Which of the following logic gates provides output as 0 when both inputs are the same (either 0 or 1)?

A) XNOR B) XOR

C) NOR D) NAND

Which of the following is not a binary number

A) 111 B) 101

C) 11E D) 000

A 3 variable Karnaugh map has _____ cells.

A) 8 B) 3

C) 16 D) 4

The Boolean function A+BC is a reduced form of

A) AB+BC B) (A+B)(A+C)

C) A'B+AB'C D) (A+C)B

If a 3-input NOR gate has eight input possibilities, how many of those possibilities will result in a HIGH output?

A) 1 B) 2

C) 7 D) 8

If a three variable switching function is expressed as the product of maxterms by f (A, B, C) = π (0,3,5,6) then it can also be expressed as sum of min terms by

A) π (1,2,4,7) B) Σ (0,3,5,6)

C) Σ (1,2,4,7) D) Σ (1,2,3,7)

Adding 1001 and 0010 gives the output of

A) 1011 B) 1111

C) 1010 D) 0000

If A and B are the inputs of a half adder, the sum is given by _____

A) A AND B B) A OR B

C) A XOR B D) A EX-NOR B

11. **A circuit that converts n inputs to 2^n outputs is called**
    A) Encoder B) Decoder
    C) Comparator D) Adder
12. **A digital multiplexer is a combinational circuit that selects**
    A) One digital information from several sources and transmits the selected one B) Many digital information and convert one
    C) Many decimal inputs and transmits the selected information D) Many decimal outputs and accepts the information
13. **The truth table for an S-R flip-flop has how many VALID entries?**
    A) 1 B) 2
    C) 3 D) 4
14. **When the J and K inputs are low, the state of the outputs Q and Q' are**
    A) Unchanged on clocking B) Changed on clocking
    C) Changed on output D) Changed on input
15. **How many select lines would be required for an 8-line-to-1-line multiplexer?**
    A) 2 B) 4
    C) 8 D) 3
16. **The characteristic equation for T flip flop is**
    A) $TQ + T\overline{Q}$ B) $\overline{T}Q + T\overline{Q}$
    C) $\overline{T}Q + \overline{T}\overline{Q}$ D) $T\overline{Q} + T\overline{Q}$
17. **The minimum number of flip-flops that can be used to construct a modulus-5 counter is _____**
    A) 3 B) 8
    C) 5 D) 10
18. **The group of bits 10110111 is serially shifted (right-most bit first) into an 8-bit parallel output register with an initial state 11110000. After two clock pulses, the register contains**
    A) 10111000 B) 10110111
    C) 11110000 D) 11111100
19. **In which of the following output will always follow a sequence either in downward or upward direction?**
    A) Counter B) Registers
    C) Both A and B D) None of these
20. **Synchronous counters eliminate the delay problems encountered with asynchronous (ripple) counters because the:**
    A) input clock pulses are applied only to the first and last stages B) input clock pulses are applied only to the stage
    C) input clock pulses are not used to activate any of the counter stages D) input clock pulses are applied simultaneo each stage

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

		Marks	BT*	CO*
1. a)	Convert the following equations into their requested canonical form (i) $X(A, B, C) = \overline{A}B + BC$ (SOP) (ii) $P(W, X, Y, Z) = (\overline{W}+X)(Y+\overline{Z})$ (POS) (iii) $F(X, Y, Z) = (X + Y)(\overline{X} + Z)$ (POS)	6	L*3	1
b)	Design a logic circuit that controls the passage of the signal 'A' according to the following requirement i) Output 'X' will equal 'A' when control inputs B and C are the same ii) 'X' will remain 'HIGH' when B and C are different	6	L4	1

c)	Using K-Map, find minimal sum expression for the following switching function. $f(a,b,c,d) = \sum m(1,3,4,5,7,8,9,11,15)$	4	L3	2	1
a)	Using QM method find the prime implicants of the function. $f(a,b,c) = \sum m(0,1,2,3,4,5,6)$	6	L3	2	1
b)	Using K-Map, find minimal sum expression for the following switching functions. (i) $f(a,b,c) = \sum m(0,2,4,6)$ (ii) $f(a,b,c) = \sum m(0,2,3,4,5,7)$ (iii) $f(a,b,c) = \sum m(1,2,3,6,7)$	6	L3	2	1
c)	Simplify the following switching equations using Boolean theorem and realize the minimal expression using basic gates. (i) $F = \bar{X}\bar{Y}\bar{Z} + X\bar{Y}Z + \bar{X}YZ + XYZ$ (ii) $F = (A + \bar{B} + C)(\bar{A} + \bar{B} + C)$	4	L3	1	1
a)	Perform the code conversion for the following and show each steps clearly. (i) $12.6532_{(10)} = (?)_2$ (ii) $110100111100.0101_{(2)} = (?)_{10}$ (iii) $CF.52_{(16)} = (?)_{10}$	6	L3	1	1
b)	For the given incompletely specified Boolean function, find the minimal sum. $f(a,b,c,d) = \sum(1,5,6,7,9,11,12,13) + \sum d(0,3,4)$	6	L3	2	1
c)	State and prove De-Morgan's theorem using the tabular method.	4	L2	1	1

Unit – II

a)	Design half subtractor and full subtractor circuits using logic gates.	6	L3	3	1
b)	Implement $f(a,b,c) = \sum m(1,4,5,7)$ using (i) 8:1 MUX (ii) 4:1 MUX using a and b as select lines.	6	L3	4	1
c)	Write short note on basic bistable element.	4	L3	4	1
a)	Explain 2-to-4-line decoder using truth table, output expression, logic diagram. Implement the given function using relevant decoder. (i) $F1 = \sum m(0,1,3)$ (ii) $F2 = \sum m(0,2,3,5,7)$	6	L3	3	1
b)	Explain the operation of master slave JK flip flop with truth table and timing diagram. Write the symbol for the same.	6	L3	4	1
c)	Design 1-bit comparator using basic gates.	4	L3	3	1
a)	With neat logic diagram and function table explain D Flip flop.	6	L3	4	1
b)	Convert JK Flip flop to (i) D Flip flop (ii) T Flip flop	6	L3	4	1
c)	Explain 4-bit parallel adder with necessary block diagram.	4	L3	3	1

Unit – III

a)	Explain the working of 3-bit ripple up counter using D flip flops.	8	L3	5	1
b)	What is a shift register? Design a 4-bit shift register using D Flipflops and explain left and right shift operations.	8	L3	5	1
a)	Design a synchronous counter to sequence 0 →2→3→6→5→1→0 using D flip flops with minimal combinational gating.	8	L3	5	1
b)	What is a Ripple counter? Design and explain 3-bit ripple down counter using T flip flops.	8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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Second Semester B.Tech (CBCS) Degree Examinations

May 2024

EC1002-2 – APPLIED DIGITAL LOGIC DESIGN

(ACT, AD, AM, CB, CC, CS, EC, EE, IS, RI, VLSI)

Duration: 3 Hours | Max. Marks: 100

Note: Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.
Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III.**

20 Marks

PART - A: MULTIPLE CHOICE QUESTIONS

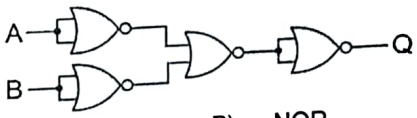
1. **Adding 1010 and 0011 gives the output of**
   A) 0111 B) 1011
   C) 1101 D) 1001

2. **Binary subtraction of 0-1 yields**
   A) Difference = 0, borrow = 0 B) Difference = 1, borrow = 0
   C) Difference = 1, borrow = 1 D) Difference = 0, borrow = 1

3. **If A and B are the inputs of a half adder, the sum is given by __**
   A) A XOR B B) A OR B
   C) A XNOR B D) A AND B

4. **The Boolean expression AB+AC'+BC simplifies to**
   A) BC+AC' B) AB+AC'+B
   C) AB+A' D) AB+BC

5. **The output of the given logic circuit represents _____ gate.**



   A) OR B) NOR
   C) AND D) NAND

6. **_____ are universal logic gates.**
   A) NAND and NOR B) NOR AND EX-OR
   C) EX-OR and EX-NOR D) NOT and AND.

7. **A 4x1 Mux will have ___ select lines**
   A) 4 B) 2
   C) 3 D) 1

8. **Which of the following expressions represent exclusive NOR of x and y?**
   A) (x xor y)' B) x xor y
   C) x or y D) x and y

9. **The truth table for an S-R flip-flop has how many VALID entries?**
   A) 3 B) 2
   C) 1 D) 4

10. **A 3 variable Karnaugh map has _____**
    A) 18 cells B) 16 cells
    C) 8 cells D) 3 cells

11. **One way to make a four-bit adder(A+B) constructed using full adders to perform subtraction(A-B) is by:**
    A) Inverting the output B) Inverting the carry-in
    C) Inverting the B inputs D) Without any changes

12. **Which is the main objective of multiplexer circuit?**
A) Decoding the binary information
B) Generation of all minterms in an output function with OR-gate
C) Generation of selected path between multiple sources and a single destination
D) Encoding of binary information

13. **If an OR gate has 4 inputs with one input high and the other three low, then the output is**
A) LOW
B) HIGH
C) HIGH or LOW depending on the input magnitude
D) Alternately HIGH and LOW

14. **In which one of the following counters, the flip flops are not clocked simultaneously?**
A) Synchronous counter
B) Asynchronous counter
C) Positive edge triggered counter
D) Negative edge triggered counter

15. **The characteristic equation for T flip flop is**
A) $TQ + T\overline{Q}$
B) $\overline{T}Q + T\overline{Q}$
C) $\overline{T}Q + \overline{T}\overline{Q}$
D) $T\overline{Q} + T\overline{Q}$

16. **If a three-variable switching function is expressed as the product of max terms by $f(A,B,C) = \pi(0,3,5,6)$ then it can also be expressed as sum of min terms by _____.**
A) $\pi(1,2,4,7)$
B) $\Sigma(0,3,5,6)$
C) $\Sigma(1,2,4,7)$
D) $\Sigma(1,2,3,7)$

17. **The characteristic equation for D flip flop is**
A) D
B) DQ
C) $\overline{D}$
D) $D\overline{Q}$

18. **A circuit that converts n inputs to 2^n outputs is called**
A) Encoder
B) Decoder
C) Comparator
D) Adder

19. **A serial in parallel out, 4-bit shift register initially contains all 1s. The data nibble 0111 is waiting to enter. After four clock pulses, the register contains _____.**
A) 0111
B) 0101
C) 1110
D) 1100

20. **Number of flip-flops required to design a counter to count from 0 to 2 is**
A) 4
B) 2
C) 6
D) 8

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

			Marks	BT*	CO*
1.	a)	Simplify the following using Boolean algebra: $(A + B' + C')(A + B' + C)(A + B + C')$.	6	L*3	1
	b)	Prove De-Morgan's theorems using the truth table approach.	6	L2	1
	c)	Minimize the given logic function using the K-map method. $F(A,B,C)=A'BC'+ABC'+ABC$.	4	L3	1
2.	a)	Given the simplified expression of a Boolean function, write the truth table, and minterm list and obtain the given simplified function first by writing the equation in canonical form and then use K-map to simplify. $Y = f(a, b, c) = c'$.	6	L3	2
	b)	Use 2's complement method to find: (i) 1101-1001 (ii) 1001-1101.	6	L3	2
	c)	Perform the conversion for the following and show each step clearly. (i) $1A2_{16} = (?)_{10}$ (ii) $45.25_{10} = (?)_2$	4	L2	1
3.	a)	A system receives three inputs and generates a one-bit output based on the even number of ones present in the input. If the inputs are completely zeros or if it has an odd number of ones, the output will be zero and if the input has an even number of ones then the output will be set to one. Write the truth table for such a system and represent the simplified function using the logic diagram.	6	L3	2

b)	Simplify the logic function f(a,b,c) =∑m(0,2,3,4,6,7) using QM technique	6	L3	2	1
c)	Use K-map to minimize the logic function f(a,b,c) =∑m(0,2,3,4,6,7).	4	L3	2	1

Unit – II

a)	Implement the given function using an active high decoder. i. F1(a,b,c)=∑m(1,2,6) ii. F2(a,b,c)= ∑ m(0) iii. F3(a,b,c)= ∑ m(3,4,5)	6	L3	3	1
b)	Convert: i) JK flip flop to D Flip Flop, ii) JK Flip Flop to T Flip Flop.	6	L3	4	1
c)	Explain the working of 4 to 2 Encoder	4	L2	3	1
5. a)	Explain with an example the working of 4-bit binary parallel adder/subtractor with relevant block diagram.	6	L2	3	1
b)	Explain the working of edge triggered D flip flop with relevant logic diagram and function table.	6	L2	4	1
c)	Explain the working of 2x1 Multiplexer.	4	L2	3	1
6. a)	Implement a Full Adder using a 4x1 multiplexer.	6	L3	3	1
b)	Explain the operation of master-slave JK flip-flop with truth table and timing diagram.	6	L2	4	1
c)	Obtain the characteristic equation of a JK flip flop.	4	L3	4	1

Unit – III

7. a)	Explain the working of a 3-bit ripple-up counter using the positive edge-triggered T flip-flops with a relevant logic diagram and the count sequence.	8	L2	5	1
b)	Explain the operations of the 4-bit serial in parallel out type shift register using D flip flops with relevant circuit diagrams and suitable examples.	8	L2	5	1
8. a)	Design a synchronous 3-bit up counter using positive edge triggered D flip flops.	8	L3	5	1
b)	Design a synchronous 3-bit down counter using positive edge triggered T flip flops.	8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Second Semester B.Tech (CBCS) Degree Examinations

May - June 2023

EC1002-1 – APPLIED DIGITAL LOGIC DESIGN

Duration: 3 Hours | Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III.**

PART - A: MULTIPLE CHOICE QUESTIONS — 20 Marks

1. **Which of the following logic gates provides output as 0 when both inputs are the same (either 0 or 1)?**
   A) XNOR B) XOR C) NOR D) NAND
2. **An XNOR gate produces an output only when the two inputs are:**
   A) High B) Different C) Low D) Same
3. **The octal equivalent of 1100101.001010 is**
   A) 624.12 B) 145.12 C) 154.12 D) 145.21
4. **Convert the binary equivalent 10101 to its decimal equivalent.**
   A) 21 B) 12 C) 22 D) 31
5. **Which of the following is not a binary number?**
   A) 111 B) 101 C) 11E D) 000
6. **What could be the maximum value of a single digit in an octal number system?**
   A) 8 B) 7 C) 6 D) 5
7. **The maximum number of bits sufficient to represent an octal number in binary is**
   A) 4 B) 3 C) 7 D) 8
8. **Convert $(22)_8$ into its corresponding decimal number**
   A) 28 B) 18 C) 81 D) 82
9. **If A and B are the inputs of a half adder, the carry is given by**
   A) A AND B B) A OR B C) A XOR B D) A EX-NOR B
10. **Half-adders have a major limitation in that they cannot _____**
    A) Accept a carry bit from a present stage B) Accept a carry bit from a next stage
    C) Accept a carry bit from a previous stage D) Accept a carry bit from the following stages
11. **If A, B and C are the inputs of a full adder then the carry is given by _____**
    A) A AND B OR (A OR B) AND C B) A OR B OR (A AND B) C
    C) (A AND B) OR (A AND B) C D) A XOR B XOR (A XOR B) AND C
12. **How many AND, OR and EXOR gates are required for the configuration of full adder?**
    A) 2, 2, 2 B) 2, 1, 2 C) 3, 1, 2 D) 4, 0, 1
13. **Let the input of a subtractor is A and B then what the output will be if A = B?**
    A) 0 B) 1 C) A D) B
14. **Let A and B is the input of a subtractor then the output will be**
    A) A XOR B B) A AND B C) A OR B D) A EXNOR B
15. **Full subtractor is used to perform subtraction of _____**
    A) 2 bits B) 3 bits C) 4 bits D) 8 bits

16. **The full subtractor can be implemented using _____**
A) Two XOR and an OR gates
B) Two half subtractors and an OR gate
C) Two multiplexers and an AND gate
D) Two comparators and an AND gate

17. **The characteristic equation for J-K flip flop is**
A) $J\bar{Q} + \bar{K}Q$
B) $JQ + KQ$
C) $\bar{J}Q + K\bar{Q}$
D) $J\bar{Q} + \bar{K}Q$

18. **The minimum number of flip-flops that can be used to construct a modulus-5 counter is**
A) 3
B) 8
C) 5
D) 10

19. **A serial in parallel out, 4-bit shift register initially contains all 1s. The data nibble 0111 is waiting to enter. After four clock pulses, the register contains _____**
A) 0111
B) 0101
C) 1110
D) 1100

20. **What is meant by parallel-loading the register?**
A) Shifting the data in all flip-flops simultaneously
B) Loading data in two of the flip-flops
C) Loading data in all four flip-flops at the same time
D) Momentarily disabling the synchronous SET and RESET inputs

PART - B: DESCRIPTIVE ANSWER QUESTIONS

		Marks	BT*	CO*
	Unit – I			
1. a)	Design a combinational logic circuit which takes two, 2 bit numbers as its inputs and generates an output to indicate when sum of the two numbers is odd.	8	L3	1
b)	Implement the following function using NAND gates only. $P = x'y + xy' + z$	4	L3	2
c)	Find the minimal sum for the following expression. $F(w,x,y,z) = \Sigma m(1,5,7,8,9,10,11,13,15)$	4	L3	2
2. a)	With a neat logic diagram, truth table and expression sketch AND, OR, NAND and EXOR gates used in digital circuits.	8	L2	2
b)	Using K-Map, find Minimal Sum expression for the following switching function $f(w,x,y,z) = \sum m(1,5,6,7,9,11,12,13) + dc(0,3,4)$	4	L3	2
c)	Perform number code conversion for the following and show each step clearly. i) $CF.52_{(16)} = ?_{(10)}$ ii) $1376.185_{(10)} = ?_{(2)}$	4	L3	1
3. a)	Using QM method, simplify the following function $f(a,b,c,d) = \sum m(0,1,3,6,8,9,12,13)$	8	L3	2
b)	Convert the following equation into Canonical SOP form $R(L,M,N) = L + \bar{M}(\bar{N}M + \bar{M}L)$	4	L3	2
c)	Define and give example to each. i) Minterm ii) Maxterm	4	L1	1
	Unit – II			
4. a)	With a neat logic diagram, function table and timing diagram explain the operation of pulse triggered JK flip-flop.	8	L2	4
b)	Implement the following functions using appropriate decoder with minimum number of gate inputs. i) $F1(w,x,y) = \sum m(1,3,5,6,7)$ ii) $F2(w,x,y) = \prod M(1,2,3,5,6,7)$	4	L3	3
c)	Differentiate between i) Combinational Logic circuits and sequential logic circuits ii) Latch and Flip flop	4	L2	4

5.	a)	Define Half and Full Adder. Design the circuit of Half and Full Adder with truth table and write the logic diagram.	8	L3	3	1
	b)	Realize the following Boolean function using an appropriate multiplexer and external gates with a, b as select lines. $f(a,b,c) = \sum m(0,1,5,6,7)$	4	L3	3	1
	c)	What are the characteristic equations of Flip flop? Obtain the characteristic equation of JK Flip Flop	4	L2	4	1
6.	a)	With neat logic diagram and function table explain the positive edge triggered D Flip flop and also write its timing diagram.	8	L2	4	1
	b)	Design a combinational logic circuit to compare the two ,1 bit numbers.	4	L3	3	1
	c)	Explain the operation of 4-bit parallel adder using full adders with an example.	4	L2	3	1

Unit – III

7.	a)	Design a synchronous counter to count $\rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 6 \rightarrow 0$ using positive edge triggered D flip flops with minimal combinational gating.	8	L3	5	1
	b)	What is a shift register? With the help of a suitable example explain the following operations: i) SIPO ii) SISO	8	L2	5	1
8.	a)	What is a Ripple counter? Explain the working of 3-bit ripple down counter with the help of a logic diagram, timing diagram and counting sequence using negative edge triggered T Flip Flops.	8	L2	5	1
	b)	Convert JK Flip flop to i) D Flip flop ii) T Flip flop	8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

First Semester B.Tech. (CBCS) Degree Examinations

December 2022

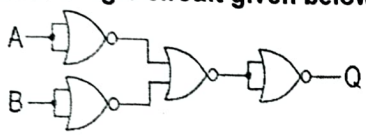
EC1002-1 – APPLIED DIGITAL LOGIC DESIGN

Duration: 3 Hours | Max. Marks: 100

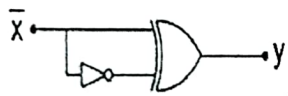
Note:
*Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.*
*Part – B: Descriptive Answer type Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II each** and **One full** question from **Unit – III.***

PART - A: MULTIPLE CHOICE QUESTIONS — 20 Marks

1. **If we add an inverter at the output of AND gate, what function is produced?**
   A) XOR B) NAND C) OR D) NOR
2. **OR gate and _____ will form the NOR gate?**
   A) OR B) NAND C) AND D) NOT
3. **What are the canonical forms of Boolean expressions?**
   A) OR and XOR B) NOR and XNOR C) MAX and MIN D) SOP and POS
4. **The Boolean expression AB+AC'+BC simplifies to**
   A) BC+AC' B) AB+AC'+B C) AB+AC' D) AB+BC
5. **The output of the logic circuit given below represents _____ gate**



   A) OR B) NOR C) AND D) NAND
6. **The output Y of the logic circuit given below is:**



   A) 1 B) 0 C) X D) X'
7. **The minimum number of NAND gates required to realise AB+AB'C+AB'C' is**
   A) 3 B) 2 C) 1 D) 0
8. **_____ are universal logic gates**
   A) NAND and NOR B) NOR AND EX-OR C) AND and NOT D) OR and EX-OR
9. **One operation that is not given by magnitude comparator is**
   A) Equal B) Less than C) Greater than D) Sum
10. **Adding 1001 and 0010 gives the output of**
    A) 1011 B) 1111 C) 1010 D) 0000
11. **Major difference between half-adders and full-adders is**
    A) Full-adders are made up of two half-adders B) Full-adders can handle double digit numbers C) Full-adders have carry input capability D) None of these

12. **Binary subtraction of 0-1 yields**
A) Difference = 0, borrow = 0 B) Difference = 1, borrow = 0
C) Difference = 1, borrow = 1 D) Difference = 0, borrow = 1

13. **How many basic binary subtraction operations are possible?**
A) 1 B) 4
C) 3 D) 2

14. **What are the two types of basic adder circuits?**
A) Sum and carry B) Half-adder and full-adder
C) Asynchronous and synchronous D) One and two's-complement

15. **Which of the following is correct for full adders?**
A) Full adders have the capability of directly adding decimal numbers B) Full adders are used to make half adders
C) Full adders are limited to two inputs since there are only two binary digits D) In a parallel full adder, the first stage may be a half adder

16. **If A and B are the inputs of a half adder, the sum is given by ____**
A) A AND B B) A OR B
C) A XOR B D) A EX-NOR B

17. **The characteristic equation for S-R flip flop is**
A) $\bar{S} + \bar{R}Q$ B) $\bar{S} + R\bar{Q}$
C) $S + \bar{R}Q$ D) $S + RQ$

18. **Which of the following statements are TRUE regarding shift registers?**
A) A shift register is a group of flip flops B) It is not used for data storage
C) It is not used for the data movement D) Shift register includes set of latches

19. **The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output register with an initial state 01110. After three clock pulses, the register contains ____**
A) 01110 B) 00001
C) 00101 D) 10101

20. **The minimum number of flip-flops that can be used to construct a modulus-5 counter is**
A) 3 B) 8
C) 5 D) 10

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

			Marks	BT*	CO*
1.	a)	Perform number conversion of the given numbers. i) $345_{10} = (?)_2$ ii) $10110.011_2 = (?)_{16}$ iii) $AB24.C5_{16} = (?)_{10}$	5	L3	1
	b)	Implement the given function using basic gates, NAND only and NOR only. Y=f(a, b, c) = a+b'+bc	5	L3	2
	c)	Design a combinational circuit having four inputs (a, b, c, d) and one output (Y). Indicate logic 1at the output when majority of its inputs are logic 1, indicate logic 0 when majority of its input are logic 0. Output is not specified when number of 1's and 0's, are equal at the input.	6	L3	2
2.	a)	Design a combinational logic circuit with inputs P, Q, R so that output S is high whenever P is zero or whenever Q=R=1.	5	L3	2
	b)	State and prove the DeMorgan's laws using truth table method.	5	L2	1
	c)	Find the minimal sum expression for the given Boolean function using QM technique. W=$f(a,b,c,d)= \Sigma m(0,1,2,3,6,7,8,9,14,15)$	6	L3	2
3.	a)	Obtain minimal sum expression for the given Boolean functions using K-map method. i) P=$f(a,b,c,d)= \Sigma m(0,1,2,3,8,9)$ ii) Q=$f(a,b,c,d)= \Sigma m(0,1,2,3,6,7,8,9,14,15)$ iii) R=$f(a,b,c,d)= \Pi M(1,2,3,4,9,10)+dc(0,14,15)$.	5	L3	2

b)	Convert the following equations into their Canonical forms. i) $R = L + \bar{M}(\bar{N}M + \bar{M}L)$ into SOP form, ii) $P = (\bar{w} + x)(y + \bar{z})$ into POS form.	5	L3	1	1
c)	Design a combinational circuit which accepts two, 2-bit binary numbers and generates three outputs. The first output indicates logic 1 whenever the 2 numbers differ by 2 or more, the second output indicates logic 1 whenever the 2 numbers are identical, and the third output indicates logic 1 when the first number exceeds second number.	6	L3	2	1

Unit – II

4. a)	Explain the operation of Master Slave JK flip flop with truth table and timing diagram. Write the symbol for the same.	5	L2	4	1
b)	Implement Y=f(a,b,c)=∑m(1,4,5,7) using i) 8:1 MUX ii) 4:1 MUX using a and b as select lines.	5	L3	3	1
c)	Design a combinational logic circuit to compare two 2-bit numbers.	6	L3	3	1
5. a)	Implement the given functions using decoder with minimum number of gate inputs. i) F1 = f(a,b) =∑m(0,1,3) ii) F2 = f(a,b,c) = ∑m(0,2,3,5,7) iii) F3 = f(a,b,c) = ∏M(1,2,3,5,6,7).	5	L3	3	1
b)	Explain 4-bit parallel subtractor with necessary block diagram.	5	L2	3	1
c)	Explain the working of +ve edge triggered D flip flop with relevant logic diagram, function table and timing diagram.	6	L2	4	1
6. a)	Design 8:3 line encoder with relevant truth table, output expressions and draw the logic diagram using basic gates.	5	L2	3	1
b)	Convert JK flip to SR and T flip flop.	5	L2	4	1
c)	Design a common cathode BCD to seven segment decoder.	6	L3	3	1

Unit – III

7. a)	Explain the working of 4-bit ripple down counter with relevant diagram and function table.	8	L2	5	1
b)	Explain the working of following shift registers with relevant block diagram and appropriate examples. i) 4-bit serial in serial out, ii) 4-bit parallel in parallel out, iii) 4-bit serial in parallel out.	8	L2	5	1
8. a)	Derive characteristic equations of JK, SR and T flip flops.	8	L2	5	1
b)	Design a synchronous counter to count 0→1→4→6→7→5→0 using positive edge triggered JK flip flops with minimal combinational gating.	8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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